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Report #3

Contribution	
Member	Contribution
AbdelRahman	EDA, Report Writing (Intro, EDA, Model)
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Table of Contents

1. Introduction & Problem Statement	3
2. Exploratory Data Analysis (EDA)	5
3. Models & Methodology	7
4. Validation & Experiments	10
5. Conclusion & Future Work.....	10
6. Lessons Learnt	12
References	13
Appendix – README for the Codes	13

List of Figures

Figure 1 Time Series Plots & Histograms of Original LMP vs. Box-Cox Transformed LMP	4
Figure 2 STL daily decomposition plots of all variables (last 30 days): (a) Top Left: STL for LMP, (b) Top Right: STL for Solar Forecast, (c) Middle Left: STL for Load Forecast, (d) Middle Right: STL for Wind Forecast, (e) Bottom Left: STL for Fuel Price, (f) Bottom Right: STL for Air Temperature	5
Figure 3 Time Series Plots (full training data) of: (a) Top plot: Solar Forecast Time Series and (b) Bottom plot: Wind Forecast Time Series	6
Figure 5 Superimposed Time Series Plot of LMP (Box-Cox $\lambda = 0.598$ Transformed) vs. LMP (Seasonal Differenced) with 24-lag	8
Figure 4 The PACF plot for the LMP training data.....	8
Figure 6 The ACF plot of the residuals from the chosen and fitted SARIMA-X model	9
Figure 7 Residual ACF Plots of 2 fitted ML Models: (a) Top: XGB Model and (b) Bottom: LGBM Model	9

List of Tables

Table 1 Resulting MAE from rolling forecasts testing of various forecasting models	10
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1. Introduction & Problem Statement

Forecasting day-ahead electricity prices is a critical challenge in the deregulated energy markets. This is due to the growing amount of variable renewable energy generation and unpredictable fuel costs over a long-time horizon [1]. Over the years, electricity prices have seen rapid proliferation of renewable energy generation, unstable fuel costs, and increasing demand patterns. The transition to a deregulated electricity system, such as in the United States, has encouraged market participants and stakeholders to optimize the bidding strategies, manage operational risk, and, thus, enhancing the grid reliability. Yet, this transition has also introduced unprecedented price variability. This limitation has given rise to the Locational Marginal Price (LMP) forecasting, which quantifies the marginal cost to deliver an additional unit of electricity (MW) at a specific node. This area of forecasting also included optimization modeling with decisions ranging from generation scheduling to demand-side bidding. Therefore, accurate LMP predictions are of importance to not only facilitate efficient market operations but also to mitigate any financial exposure of utilities and energy generations.

In our initial report (Report #1), we implemented a multi-stage forecasting pipeline, where with exploratory data analysis (EDA) we uncovered key seasonal cycles and covariate relationships; the unknown covariate in the testing set, fuel price, were predicted using a naïve forecast; and a seasonal ARIMA with exogenous regressors (SARIMA-X) modeling was introduced to regress and integrate the load, solar, wind, fuel price, and temperature covariate predictors into the LMP forecasts. This forecasting pipeline laid the foundation of our methodology by identifying a dominant 24-hour seasonality, linear correlations between LMP and predictors—positive for load and fuel price, inverse for solar and wind—and weekend effects, which motivated the idea of dummy variable inclusion. However, the residual diagnostics revealed some instability in the model, where the ACF plot showed that there is a significant pattern uncaptured by the model.

Building on this foundation and insights, the second report (Report #2) focused mainly on model refinement and parameter tunings. Rather than using R automated pick for the SARIMA-X parameters, we adopted a SARIMA(3,0,2)(0,1,2)[24] specification with no intercept and using stepwise estimation and likelihood approximations to reduce the computational time below 1,000 seconds. This LMP forecasting model achieved a three-day ex-post MAE of 3.78, which outperforms the naïve benchmark of 5.10. In addition, iterative experiments tested lagged regressors, Fourier terms, and transformations. Yet, we found that attempts at seasonal differencing of regressors and Fourier terms yielded some marginal gains in the model but at the expense of complexity and sometimes degraded residual behavior. These efforts made in the second report highlight the trade-off between enhancing model reliability without inflating run-time or overfitting the model with noisy patterns.

Despite the advances in the model, still two pivotal challenges persist in the day-ahead LMP forecasting. The first challenge is the inherent heteroscedasticity of electricity prices, where the variance of the residuals in the regression modeling is not constant. This is seen by the sudden

spikes and variability that defy the constant variance assumptions of the SARIMA-X model. The second challenge is the non-Gaussian distribution of the residuals. This can bias the parameter estimations and weaken prediction accuracy. In practice, addressing these issues is critical for both improving point forecasts and probabilistic assessments. In classical regression time series contexts, transformations, such as Box-Cox, help to stabilize the variance by inducing homoscedasticity and normality in the time series.

To tackle these limitations, our final report presents the same SARIMA-X pipeline modeling, but with incorporating a Box-Cox transformation. In the final code, we computed the Box-Cox parameter $\lambda \approx 0.598$ using Guerrero's method on the full training set to ensure that positive data

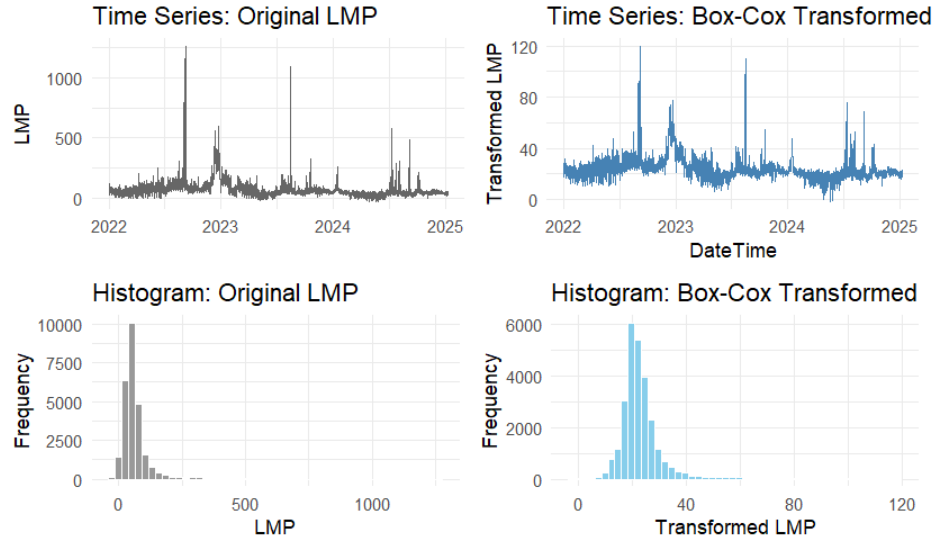


Figure 1 Time Series Plots & Histograms of Original LMP vs. Box-Cox Transformed LMP

requirements are met. The Guerrero's method partitions the time series into its seasonal cycles, computes mean and variance for each of the non-overlapping seasonal period after applying a candidate Box-Cox transformation, then selects the best λ that minimizes the coefficient of variate of those period means. Post-transformation, as observed in the histogram of the transformed series figure 1, the LMP series exhibits less skewness and stabilized variance. Then, the transformed series is inputted into the SARIMA(3,0,2)(0,1,2)[24] fitting model with five external regressors and bias adjustments.

Problem Statement. While the SARIMA-X model effectively captures daily patterns and external covariates effects, the forecast accuracy and reliability are limited due to the remaining variance heterogeneity and non-Gaussian residuals behavior. The two core problems addressed in this report are: (1) Variance stabilization. A variance stabilizing transformation (in our case, Box-Cox) is used to align the residual behavior with Gaussian error structures and improve parameter estimation. (2) Model parsimony vs. reliability. Engineered covariates such as dummy variables, lagged regressors, or seasonal differencing can improve the model fit but risk overfitting and increased run-time. Therefore, adding Box-Cox λ estimation must preserve the model parsimony, while strengthening the forecast stability.

Building on the problem statement and the proposed Box-Cox augmented SARIMA-X pipeline, this section presents the key characteristics of the data that motivated each modeling choice. We begin by examining LMP series with seasonal decomposition. Then, we explore the relationships between LMP and each predictor through scatter plots. Finally, we analyze the inherent autocorrelation structures to justify the chosen seasonal orders.



Figure (2a) presents the daily STL (Seasonal-Trend decomposition using Loess) of the LMP series, separating the trend, seasonal, and residual components for the last 30 days in the train set (from 2024-12-09 19:00:00 to 2025-01-08 19:00:00). The seasonal component exhibits a string 24-hour cycle. This justifies the need for a seasonal period of 24 in the SARIMA specification. The residual component shows an irregular distribution that violates the constant-variance assumptions, thus motivating the Box-Cox transformation pipeline. Figure (2c), STL decomposition of load covariate, shows a consistent daily pattern, suggesting a strong performance when treated as an exogenous series. Both solar (2b) and wind (2d) covariates exhibit a clear daily cycle but also abrupt shifts in amplitude over time. Thus, their intraday patterns support the inclusion via deterministic seasonal regressors. The fuel price series (2e) is relatively smooth, with low-frequency trend changes. Its relatively stable variance component contrasts with the LMP's variance. Indicating the importance of including this as a regressor. The air temperature time series (2f) shows a sinusoidal seasonal component at the daily and yearly scale. Although its direct correlation with LMP is weaker than that of the other covariates, the physical demand response to temperature extremes (heating/cooling) justifies its inclusion.

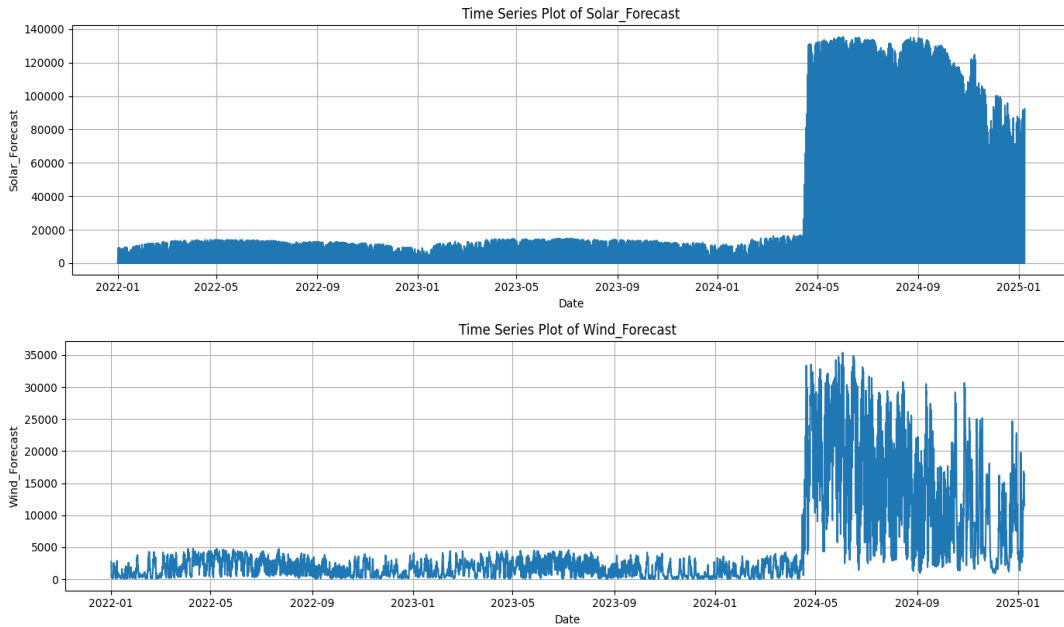


Figure 3 Time Series Plots (full training data) of: (a) Top plot: Solar Forecast Time Series and (b) Bottom plot: Wind Forecast Time Series

As seen in the time series plots in figure 2, both the solar (3a) and wind (3b) exhibit a sudden, significant increase, roughly ten-fold jump in magnitude, beginning in mid-April/early May 2024. This reflects a change in the upstream data scaling rather than a genuine surge in renewable output, as the increase magnitude is not justifiable. To prevent this structural change from affecting the Box-Cox SARIMA-X estimation, we conditionally rescale all post-April 13, 2024, values by dividing them by 10 within both the training and testing sets. Justification to this transformation is to restore continuity in the covariate scales to preserve the integrity of the λ estimation.

3. Models & Methodology

For LMP modeling, the primary factors for model selection are the strong seasonality and presence of insightful covariates, which motivate the implementation of models that can handle both aspects of the data. There are two main classifications of models that fit this description: statistical models and machine-learning (ML) models. Under the statistical modeling umbrella are the SARIMA-X models and the harmonic regression models. ML models consist of an even broader spread of models, but the ML models that are evaluated for the purpose of this study are the Extreme Gradient Boost (XGB) and Light Gradient Boosting Model (LGBM), both tree-based gradient descent boosting models that are known to be able to capture non-linear relationships in data while incorporating exogenous variables. Prior to forecasting the LMP, however, the fuel price must be forecasted since person-correlation analysis revealed how impactful it is for determining the resulting LMP, and, unlike the other exogenous variables, the forecast is not provided ahead of time. Historical analysis of the fuel price reveals relative stability and constant daily behavior. Furthermore, in the 30 days leading up to the most recent testing window, the fuel price only exhibited a range of \$1.74 with a standard deviation of 0.42. This, along with the fact that the forecast horizon is only one day, motivates the use of the naïve forecast for the fuel price.

For LMP modeling, the aforementioned statistical and ML models are implemented along with the naïve and seasonal naïve models to serve as a benchmark. To train the models, data from January 1st, 2023, through December 1st, 2024, is used, while the last month of the training data (after December 1st, 2024) is reserved for rolling testing to validate the model performance. The decision to exclude data from 2022 was made after comparing residual analysis performed on models trained with both the entire data set as well as the shortened set. The abbreviated data set led to improved results on most models in terms of both residual ACFs and rolling testing with the reserved training data (see the following section for more details). With the choice of training data solidified, the choice of modeling becomes easier. We once again rely on residual analysis conducted during training to evaluate model performance since the true values for the testing set are unknown. Ideal residuals resemble white noise with a relatively constant variance, zero mean, and Gaussian distribution, which would indicate that the model has captured all of the relevant information. To assess this criterion, we examine the residual time series plot, histogram and ACF plot.

Starting with the statistical models, the dynamic harmonic regression model did not produce good results at all, despite the inclusion of lagged regressors and trigonometric features to capture daily seasonality. The residuals exhibit significant autocorrelation at all relevant lags. The next statistical model tested and ultimately chosen to be our final forecast model is the SARIMA-X model. Specifically, a SARIMA(3,0,2)(0,1,2)[24] with exogenous regressors. The following sections will dive deeper into the experiments that support this decision, but from a scientific perspective, there are a few reasons this model is chosen. First, the obvious daily seasonality brought about by environmental factors leads to the seasonal period of 24 hours

being chosen. Then, a seasonal difference is applied to stabilize the variance and make the time series more stationary, which is known to improve model performance. Although the LMP series is relatively stationary, it demonstrates some volatility and cyclical behavior, which the seasonal differencing largely corrects. Figure 4 shows the comparison between the LMP (post box-cox transformation) with and without seasonal differencing.

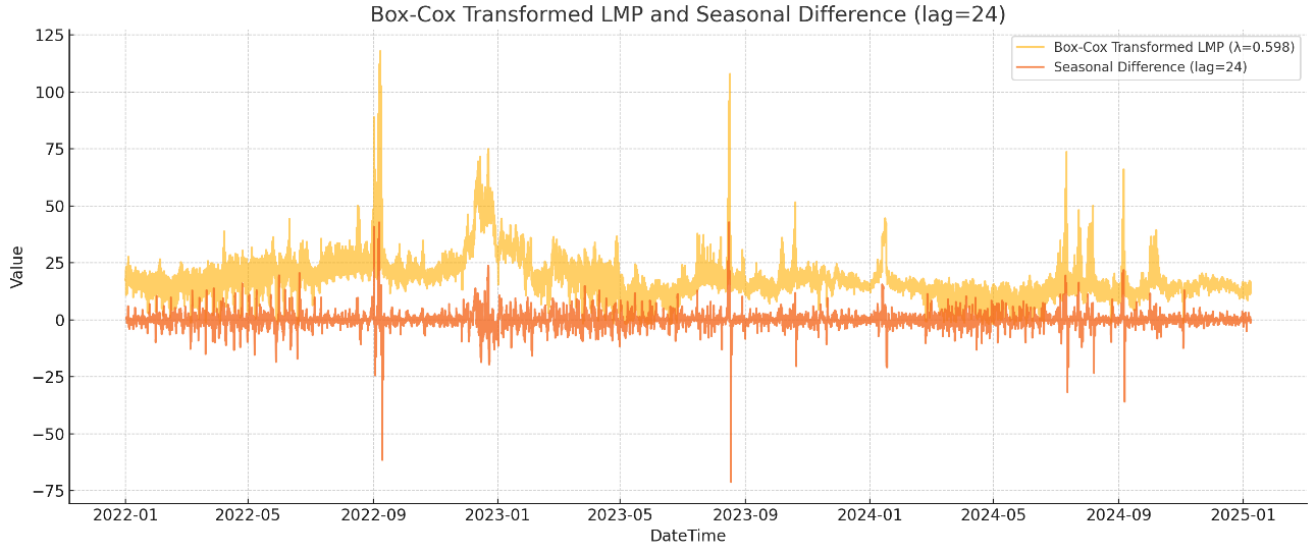


Figure 4 Superimposed Time Series Plot of LMP (Box-Cox $\lambda = 0.598$ Transformed) vs. LMP (Seasonal Differenced) with 24-lag

One can clearly see that the differenced data is more stationary, making it more suitable for forecasting. Another key aspect of the SARIMA-X model selection is the inclusion of three autoregressive terms. Figure 5 shows the results of the partial autocorrelation function (PACF) analysis of the LMP training data. It can clearly be seen that the third lag is the last

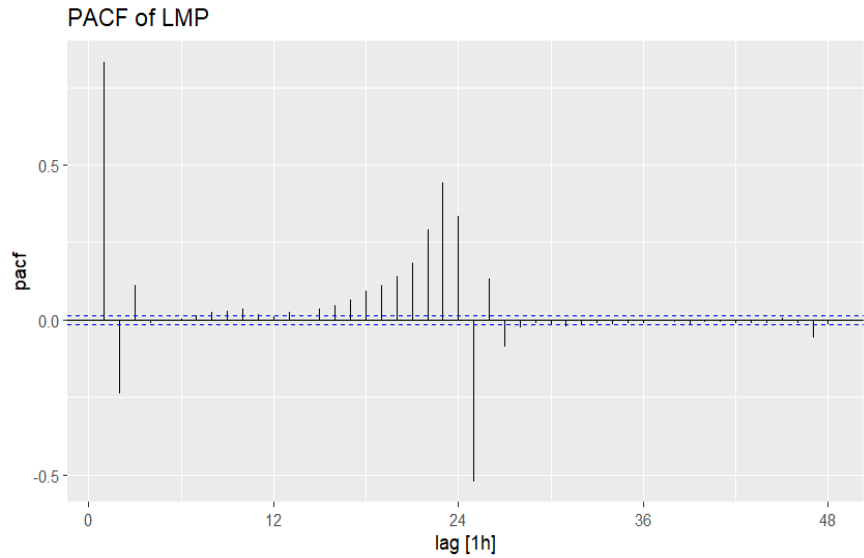


Figure 5 The PACF plot for the LMP training data

significant lag for LMP, outside of the lags later in the period corresponding to seasonality. As a result, an $AR(p = 3)$ is chosen for the number of autoregressive terms in the model. Finally, the moving average terms, which capture the short-term dependencies in forecast errors, are chosen based on experimentation and analysis of the residuals, to be discussed in the following section.

As for the regressors, all five that are provided (Solar and Wind Forecast, Air Temperature, Fuel Forecast, and Load Forecast) are incorporated into the model as they all demonstrate at least modest cross-correlation with LMP and help capture features like seasonality. Figure 6 shows the resulting residual ACF plot for the SARIMA-X demonstrating very strong results.

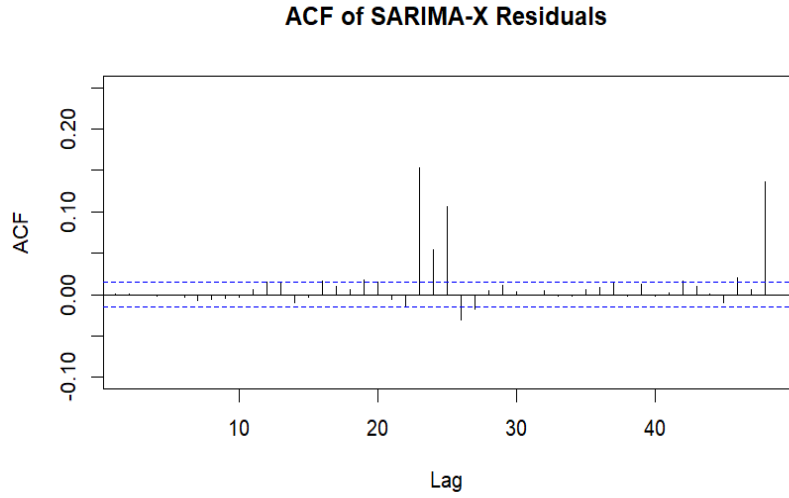


Figure 6 The ACF plot of the residuals from the chosen and fitted SARIMA-X model

Finally, the XGB and LGBM models are chosen to serve as benchmark or possible replacement models for the chosen SARIMA-X model, should they perform better. ML models, however, require more extensive feature engineering because models like XGB do not inherently account for temporal dependencies and patterns present in time series data¹. Accordingly, in addition to the features already present in the SARIMA-X model, temporal features like month, week, and hour, and even trigonometric features to help detect seasonality, are included as inputs for training. Additionally, parameter optimization is performed using a grid search method to best fit the training data. From a residual analysis perspective, both the XGB and LGBM performed similarly and quite adequately, with both of their ACF plots passing the Ljung–Box Test, but some signal remained, and ultimately, as will be discussed in the following section, the SARIMA-X model performed better. Figure 7 shows the residual ACF plots for the two ML models, and while most ACFs fall below the line of significance, there appears to be a significant signal remaining at lags of hours 1 and 2, as well as multiples of 24, suggesting improvement is possible.

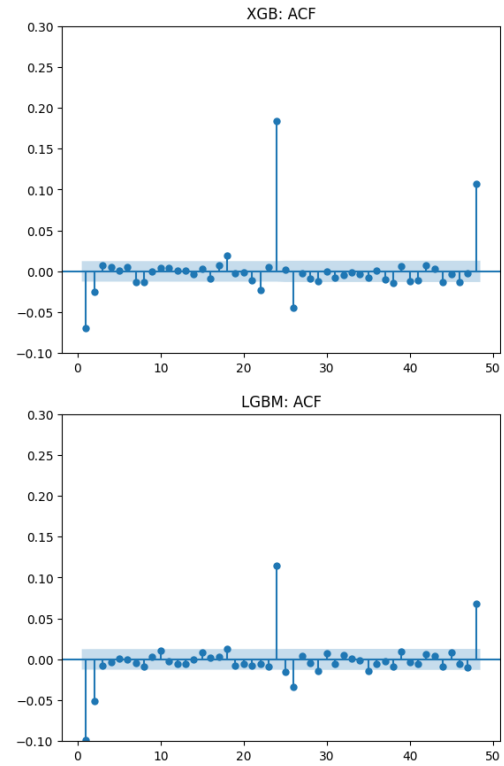


Figure 7 Residual ACF Plots of 2 fitted ML Models: (a) Top: XGB Model and (b) Bottom: LGBM Model

4. Validation & Experiments

To validate and test our models, we implement a 24-hour rolling testing approach performed over a period of 37 days (since Dec. 1st) that tests the trained model on data it has not yet seen, and the mean absolute error (MAE) is computed to evaluate performance. MAE is the chosen evaluation metric because it is the judgment metric for the competition. Due to computational constraints (the SARIMA-X model takes ~ 16 minutes to train), the model is not retrained each day it makes a forecast during rolling testing and instead is just given updated exogenous inputs. Rolling testing is performed on the SARIMA-X model, the ML models, as well as the naïve and seasonal naïve benchmarks for comparison. Table 1 shows the resulting MAEs for the previously discussed models.

Table 1 Resulting MAE from rolling forecasts testing of various forecasting models

Forecast Model	MAE	% MAE Improvement on Naïve
Naïve	5.79	N/A
Seasonal Naïve	4.28	26.08%
XGB	3.98	31.26%
LGBM	4.02	30.57%
SARIMA-X	3.09	46.65%

The results from the rolling test, along with residual diagnostics, clearly show that the proposed SARIMA-X model demonstrated superior performance to all of the benchmark models. While the ML models did produce an improvement over the benchmark Naïve models, the SARIMA-X model scored an additional 15% improvement over the naïve forecast, highlighting its strong performance.

5. Conclusion & Future Work

In this project, we utilized the SARIMA-X regression formalism to fit a model that is complex enough to correctly represent and forecast LMP values with a good accuracy. In addition, we extended and enhanced this framework by integrating a Box-Cos transformation ($\lambda \approx 0.598$ using Guerrero’s method) to stabilize the variance and normalize the residuals. When retrained on the full dataset, the Box-Cox-SARIMA-X model achieved a one-day-ahead test MAE of 2.2643. Also, with a 37-day rolling evaluation, the same regression pipeline yielded an MAE of 3.0923. This demonstrates that the performed transformation did preserve the forecast accuracy while tackling the heteroscedastic spikes. These results and the MAE comparison in table 1 confirm that using a single SARIMA(3,0,2)(0,1,2)[24] model, augmented by five regressors and transformed for variance stability, can capture both the cyclical and unstable dynamics in day-ahead LMP forecasting.

The revised forecasting pipeline proceeds as follows:

1. Using Guerrero's method in Box-Cox estimation, determine λ on the training LMP to optimize the transformation towards normality and homoscedasticity.
2. Fit the Box-Cox transformed LMP series with exogenous covariates using maximum likelihood SARIMA-X, while holding the fuel price interpolation constant via naïve forecasts.
3. Perform a rolling 24-hour forecast workflow on post-December data period by feeding the model new observations without refitting its coefficients (so the same parameter estimates are used each day).
4. Retrain on the entire test set with the established λ , then forecast the test set and compute the MAE.

Future work:

1. Evaluate a fully probabilistic forecasts by implementing and comparing interval scoring metrics such as Quantile Scores (QS), Winkler Scores (WS), and Continuous Ranked Probability Score (CRPS) to assess both calibration and sharpness of the LMP prediction intervals.
2. Develop a rolling or regime-specific Box-Cox transformation to the LMP time series by estimating λ using maximum likelihood over the sliding window or within detected price regimes. This will result in a stabilized variance around the structural breaks and seasonal shifts.
3. Incorporate Dynamic Harmonic Regression modeling using Fourier terms to capture the intraday and intraweek seasonal patterns in the real-time LMP time series before fitting them to a model. However, this has to be done efficiently, as in the case of SARIMA-X models, adding Fourier terms may result in a lower prediction accuracy and longer model fitting run time.
4. Compare the performance of the SARIMA-X model with a VARIMA-X multivariate model (jointly modeling several interconnected LMP nodes with exogenous predictors, specifically load and renewables, to develop spatial and cross-series correlations).
5. Explore other advanced machine learning approaches, such as gradient boosting, random forest, or deep neural networks, coupled with feature engineering, e.g., lagged values, interaction terms, on non-linear transforms, as the main modeling pipeline to fully capture the complex patterns in LMP dynamics.

6. Lessons Learnt

This month-long project challenged us in unexpected ways and forced us to adapt our original thinking in more ways than one. Although many of these lessons were covered in lecture, it wasn't until we had to implement them ourselves that they truly resonated. The hands-on application helped solidify our understanding in a way that theory alone could not. The biggest lesson learnt is that there is a large tradeoff between model complexity and training time. Even the relatively simple SARIMA-X model that we chose took over 15 minutes to train so even to make a small adjustment, we would have to wait at least that amount of time to see if the changes paid off. At times, this was discouraging, especially if the changes produced worse results, but it led us to the next lesson we learned: the importance of organization and strategy. Since the wrong modification to a model could lead to the waste of nearly an hour, it was important that we followed the scientific methodology that we learned in class when it comes to model selection and modification. The phrase "there's not enough time to try everything" didn't resonate with us until it took over an hour to try to fit a dynamic harmonic regression model. The next lesson we learned was to be cautious of forecasting based on a forecast. In the early stages of the project, we fit a large number of more dynamic fuel price models, but this led to wild forecast swings that were highly dependent on the accuracy of the said fuel price forecast. For this reason, we chose to stick with the naïve forecast and mention that the forecast accuracy of our model is contingent on the previous day's fuel price. One other significant lesson I learnt that I'll mention is the new experience we gained with ML models. As we alluded to in the report, we tried to use the same forecast inputs that we used for the SARIMA-X model for the ML models and had surprisingly poor results (performance was comparable to the naïve forecast). After doing some brief research, we learned that we need to do much more significant feature engineering for ML models to capture more of the signal in the data. An interesting observation, however, is that even with the substantial feature engineering, the ML models trained in significantly less time (seconds as opposed to minutes), which allowed for much more experimentation in the short period of time that we became aware of this type of model class. Overall, this project not only deepened our understanding of forecasting theory but also taught us the practical discipline, strategic thinking, and adaptability required to apply that theory effectively in real-world scenarios.

References

- [1] Weron, R. (2014). *Electricity price forecasting: A review of the state-of-the-art with a look into the future*. International Journal of Forecasting, 30(4), 1030–1081.
<https://doi.org/10.1016/j.ijforecast.2014.08.008>

Appendix – README for the Codes

```
library(forecast)
library(readxl)
library(lubridate)
library(dplyr)
library(ggplot2)
library(tibble)
# === 1. Read and preprocess training data ===
train_data <- read_excel("C:/Users/nperc/Downloads/Train_Round8.xlsx") %>%
  filter(DateTime > as_datetime("2023-01-01", tz = "America/Los_Angeles")) %>%
  mutate(DateTime = as_datetime(DateTime, tz = "America/Los_Angeles")) %>%
  mutate(
    Solar_Forecast = if_else(DateTime > as_datetime("2024-04-13", tz =
"America/Los_Angeles"),
      Solar_Forecast / 10, Solar_Forecast),
    Wind_Forecast = if_else(DateTime > as_datetime("2024-04-13", tz =
"America/Los_Angeles"),
      Wind_Forecast / 10, Wind_Forecast)
  )
# === 2. Read and preprocess test data ===
test_data <- read_excel("C:/Users/nperc/Downloads/Test_Round8.xlsx") %>%
  mutate(DateTime = as_datetime(DateTime, tz = "America/Los_Angeles")) %>%
  mutate(
    Solar_Forecast = if_else(DateTime > as_datetime("2024-04-13", tz =
"America/Los_Angeles"),
      Solar_Forecast / 10, Solar_Forecast),
    Wind_Forecast = if_else(DateTime > as_datetime("2024-04-13", tz =
"America/Los_Angeles"),
      Wind_Forecast / 10, Wind_Forecast)
  )
# === 3. Compute Box–Cox  $\lambda$  for LMP ===
lambda_lmp <- BoxCox.lambda(train_data$LMP, method = "guerrero", lower = 0)
# === 4. Prepare exogenous variables (forecast naive fuel price) ===
xreg_train <- as.matrix(train_data %>%
  select(Load_Forecast, Solar_Forecast, Wind_Forecast, Fuel_Price, AirTemp_C))
test_exo <- test_data %>%
  select(Load_Forecast, Solar_Forecast, Wind_Forecast, Fuel_Price, AirTemp_C)
mis_idx <- which(is.na(test_exo$Fuel_Price))
```

```

if (length(mis_idx) > 0) {
  fm <- naive(ts(train_data$Fuel_Price, frequency = 24), h = length(mis_idx))
  test_exo$Fuel_Price[mis_idx] <- as.numeric(fm$mean)
}
xreg_test <- as.matrix(test_exo)
y_test <- test_data$LMP
# === 5. Fit SARIMA-X on full training data (~16 minutes) ===
y_train <- ts(train_data$LMP, frequency = 24)
sarimax_model <- Arima(
  y = y_train,
  xreg = xreg_train,
  order = c(3, 0, 2),
  seasonal = list(order = c(0, 1, 2), period = 24),
  include.constant = FALSE,
  lambda = lambda_lmp,
  biasadj = TRUE,
  method = "ML"
)
# === 6. Forecast on test set ===
f_test <- forecast(sarimax_model, xreg = xreg_test, h = nrow(test_data),
  lambda = lambda_lmp, biasadj = TRUE)
preds <- as.numeric(f_test$mean)
# === 7. Export results ===
test_df <- tibble(
  DateTime = test_data$DateTime,
  Forecast = preds,
  Actual = y_test,
  Error = abs(preds - y_test)
)
# === 8. Display results ===
ggplot(test_df, aes(x = DateTime)) +
  geom_line(aes(y = Actual, color = "Actual")) +
  geom_line(aes(y = Forecast, color = "Forecast")) +
  labs(title = "SARIMA-X Forecast vs Actual LMP",
    x = "DateTime", y = "LMP ($/MWh)") +
  scale_color_manual("", values = c("Actual" = "black", "Forecast" = "blue")) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```